

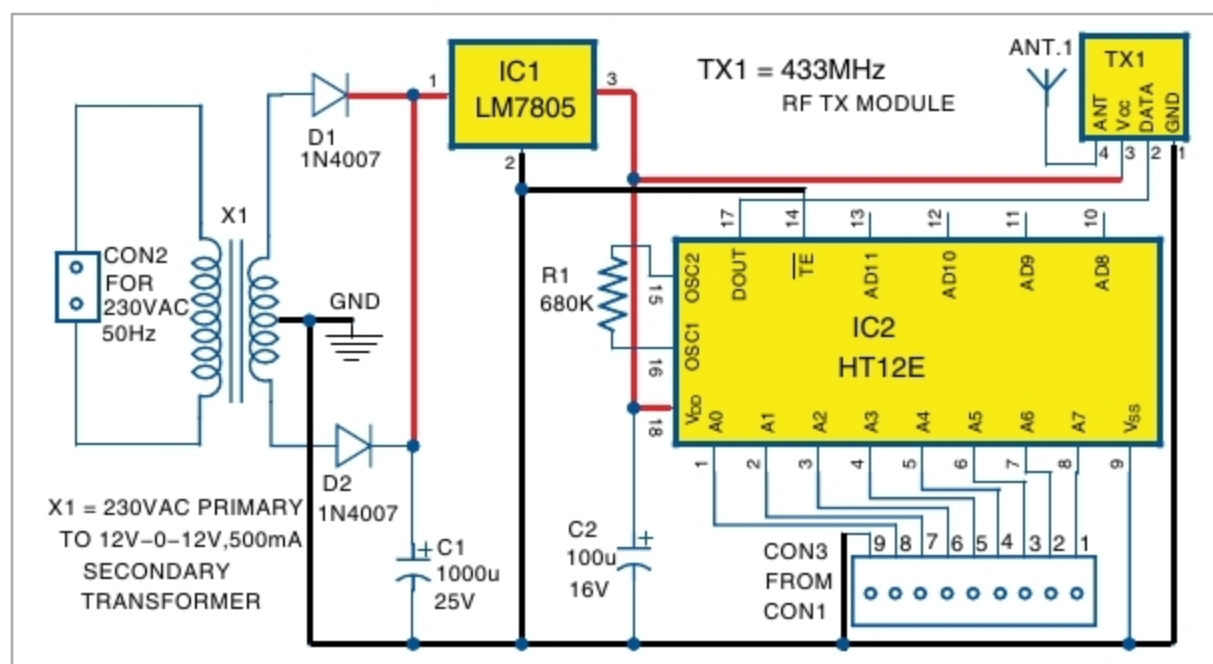


Fig. 3: Circuit diagram of the transmitter unit

divided into two units, transmitter and receiver.

Transmitter unit. Fig. 3 shows the schematic of the transmitter unit of the water-level controller. It comprises sensor assembly, 5V voltage regulator 7805 (IC1), encoder HT12E (IC2) and 433MHz RF transmitter module (TX1). Transformer X1 converts 230V AC to 12V-0-12V, which is rectified by the diodes (D1 and D2) and regulated by IC1 to provide regulated 5V. Capacitors C1 and C2 are used for filtering. 5V supply enables the circuit.

The sensor assembly holds eight metal strips (S0 through S7) clamped over a plastic pipe (or insulated metal pipe), which is erected vertically from the bottom of the tank. The metal strips act as water-level sensors (Fig. 4). These are generally spaced equally along the length of the pipe from the bottom to full tank height, at 1/8th, 2/8th, etc positions of the pipe length. (These can also be spaced at unequal distances, if desired.) Wires run from

these strips to the address lines of the encoder IC. The lower most one (S0) goes to A0 (pin 1) of the encoder (IC2). The next sensor (S1) goes to A1 (pin 2) and so on, up to S7 that goes to pin 8 of the encoder.

An additional metal strip SG, clamped over the pipe at the bottom of the tank, goes to circuit ground. In case of a metal tank, the tank itself can act as SG. TE input (pin 14) of the encoder is permanently grounded. So, the encoder is configured to produce encoded data continuously. Data is available at its DOUT line (pin 17).

First, connect the sensors from the tank to the circuit and power on the circuit. If water level in the tank stays below

S0, no sensor is grounded by conductive water and, hence, digital address input to the encoder is 11111111. If water is pumped into the tank, the level starts rising. When it reaches S0, digital address becomes 11111110. If water level keeps increasing, address lines of encoder HT12E follow digital 8-bit numbers, as tabulated in Table 1.

When tank is empty, address line of encoder will be 11111111. When it is full with water-level encoder address will be 00000000.

Hence, at any instant, digital number so produced by the sensors and 4-bit data (if any) at data input lines (pins 10, 11, 12 and 13) of the IC are encoded to serial form by the IC. Encoded serial data, available from pin 17 of HT12E, is steered to data input pin of 433MHz TX module. Module operating in ASK/OOK mode transmits the same to a remote receiving end, wirelessly.

Receiver unit. Fig. 5 shows the schematic of the receiving unit of the water-level indicator-cum-controller. It is built around transformer X2, 5V voltage regulator 7805 (IC8), NAND gate 4011 (IC3), TTL NAND gate

Address Lines of Encoder HT12E When Water Level Increases								
Sl. No.	A7	A6	A5	A4	A3	A2	A1	A0
Initial status	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	0
2	1	1	1	1	1	1	0	0
3	1	1	1	1	1	0	0	0
4	1	1	1	1	0	0	0	0
5	1	1	1	0	0	0	0	0
6	1	1	0	0	0	0	0	0
7	1	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0

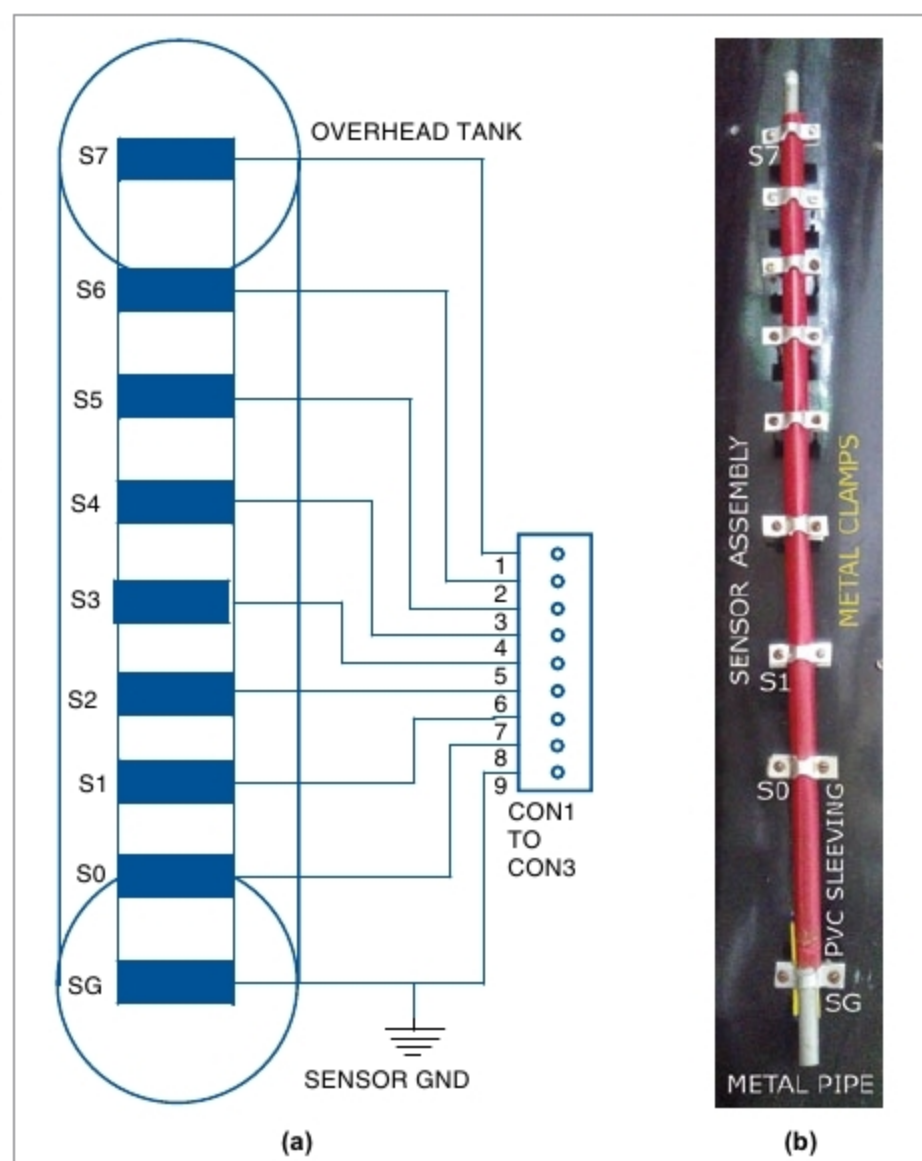


Fig. 4: Assembly diagram (a) and actual assembly (b) of the sensor